NCCP 00722 · VERSION 2

Your treatment, explained

HER2-positive early breast cancer · before surgery · NCCP 00722 v2 · Document revised 4 November 2026

> **Your personal plan:** this is the standard regimen schedule. Your team confirms your prescribed doses, cycles, treatment changes and appointments in your written plan.

TCHP: docetaxel, carboplatin, trastuzumab and pertuzumab

21-day cycle

Source reconciled; independent clinical, pharmacy and patient/PPI review pending

ACT ON SYMPTOMS, NOT THE CALENDAR

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathlessness, chest symptoms, severe diarrhoea, dehydration, jaundice, unusual bleeding or sudden illness promptly. Do not wait for the next planned test.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

Open your CUH contact / treatment card

Sources: nccp · hse

YOUR COMPLETE REFERENCE

Read what you need, when you need it.

Start with the overview, explore the cycle, or go straight to a question. Urgent advice is always available.

- How each drug works
- The phases and how they connect
- Your 21-day timeline
- Baseline and cycle-specific checks
- Whole treatment, scans and checks
- Tablets, visits and practical planning
- Why this treatment?
- Your treatment and the next step
- When scans, heart checks and bloods happen
- How surgery and radiotherapy fit
- Supportive medicines
- Questions to ask
- Sources

UNDERSTAND YOUR TREATMENT

What each medicine does

Treat HER2-positive early breast cancer as part of a plan intended to reduce the risk of it returning.

TAXANE CHEMOTHERAPY

Docetaxel



Disrupts the cell’s division machinery

Docetaxel stabilises tiny structures called microtubules. Cells need these to separate their chromosomes during division; disrupting the process can stop cancer cells multiplying. Healthy dividing cells are also affected.

Microtubules stabilised → Division disrupted → Cell growth affected

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What to watch for

Take prescribed steroid premedication. Report fever, lasting numbness, swelling, sore nails or infusion symptoms.

Sources: nccp

PLATINUM CHEMOTHERAPY

Carboplatin



Damages DNA inside cells

Carboplatin creates links in DNA, making it harder for cells to copy and repair their genetic instructions. Its dose depends on kidney function as well as the intended regimen.

Platinum enters cell → DNA links form → Copying is disrupted

What to watch for

Keep blood-test appointments. Fever, unusual bleeding, repeated vomiting or infusion symptoms need prompt advice.

Sources: nccp

HER2-TARGETED ANTIBODY

Trastuzumab



Targets HER2 growth signals

Trastuzumab binds to the HER2 protein on cells and helps reduce HER2-driven growth and recruit immune activity. HER2 testing identifies the cancers for which this approach is used.

HER2 on cell surface → Antibody binds → Growth signalling reduced

What to watch for

Attend planned echoes. Report breathlessness, ankle swelling, rapid weight gain or infusion symptoms.

Sources: nccp

HER2-TARGETED ANTIBODY

Pertuzumab



Blocks HER2 pairing with other receptors

Pertuzumab binds a different area of HER2 from trastuzumab. It limits receptor pairing, so the two antibodies act together at different points on the same growth pathway.

HER2 pairing → Pertuzumab blocks pairing → Growth signals reduced

What to watch for

Report diarrhoea early, especially with poor fluid intake. Keep cardiac checks and report infusion symptoms immediately.

Sources: nccp

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Explore your existing anatomy and side-effect map

THE BIGGER TREATMENT PLAN

The phases and how they connect

21-DAY CYCLE

Before surgery · six TCHP cycles



All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump.

INDIVIDUAL PLAN

Surgery and selected postoperative treatment



After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

Relative sequence, not equal lengths or booked dates. Your team confirms actual appointments and the active medicines.

Sources: nccp

BASELINE → TREATMENT → FOLLOW-UP

Tests and milestones through treatment

Columns show sequence, not equal durations or appointments. A test is only complete when performed and reviewed. Echo timing and cancer-imaging timing are different. Symptoms can bring any check forward.

Check	Before starting	Cycles 1–4	Around cycle 5 / week 12	Surgery / next phase	Ongoing / follow-up
Blood tests	Baseline FBC, renal and liver profile	Before each cycle; additional dose checks as prescribed	Continue the required pre-dose checks	Use the selected active regimen's blood-test schedule	Investigate symptoms / persistent abnormalities
Echo / cardiac assessment	Baseline cardiac assessment	Earlier for symptoms / concern	Echo during neoadjuvant treatment, around cycle 5	Surgery does not restart the echo clock	Agree follow-up for the active medicine and cardiac risk
Cancer imaging / markers	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Before surgery / pathology	Breast surveillance after early treatment
Surgery / radiotherapy	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Endocrine / bone health	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	DEXA when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00722 v2 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

TCHP: docetaxel, carboplatin, trastuzumab and pertuzumab: the whole pathway

HER2-positive early breast cancer · before surgery

All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

On a small screen, swipe the map sideways. The checks below give the same timing in words.

Treatment, tests and reviews across the course

Care / check	Before starting	Cycles 1–4	Around cycle 5 / week 12	Surgery / next phase	Ongoing / follow-up
Cancer medicines NCCP	Confirm diagnosis and exact regimen	See the dose-day map: use the current regimen and active phase	Echo due; treatment review	Surgery/pathology select postoperative medicine	Review benefit / tolerance and next plan
Surgery / radiotherapy EARLY	Agree indication and sequence	No automatic concurrent radiation	Coordinate with both teams	Local treatment as indicated	Review recovery and late effects
Blood tests NCCP	Baseline FBC, renal and liver profile	Before each cycle; additional dose checks as prescribed	Continue the required pre-dose checks	Use the selected active regimen's blood-test schedule	Investigate symptoms / persistent abnormalities
Echo / cardiac checks NCCP · ESC	Baseline cardiac assessment	Earlier for symptoms / concern	Echo during neoadjuvant treatment, around cycle 5	Surgery does not restart the echo clock	Agree follow-up for the active medicine and cardiac risk
Scans / tumour markers EARLY	Baseline disease assessment	Clinical response; imaging if concern	Response review as planned; not tied to echo	Before surgery / pathology	Breast surveillance after early treatment
Endocrine / DEXA BONE	Check hormone receptors / fracture risk	Plan endocrine therapy if indicated	DEXA when relevant to bone-risk plan	Endocrine timing by the treating team	Individual DEXA / bone follow-up

What each check is for

1

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.

Check safety before the next dose.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

Sources: NCCP

2

Heart tests

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Monitor pumping function and relevant rhythm risk.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

Sources: NCCP · HERCEPTIN · ESC

3

Cancer imaging

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine CT, PET or tumour-marker testing is not the usual screening plan for someone without symptoms.

Assess response or follow-up needs.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

Sources: EARLY

4

Bone health

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

Assess fracture risk where relevant.

Ask / act: Agree the booking owner, result reviewer and action if overdue.

Sources: BONE

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Sources checked 2026-10-21 · Source reconciled; independent clinical, pharmacy and patient/PPI review pending · SACTCheck v0.84.16

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump.

Day of cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
All regimen infusions · Day 1	IV																				
Blood counts / infection	Tests and symptoms guide review																				
Energy and bowel symptoms	Timing varies; report changes early																				
Numbness / tingling	Can accumulate with taxane treatment																				
Heart symptoms	Any time; attend planned echoes																				

Planned treatment: All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump.

Next phase: After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

Heart checks: Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

> **Why these bands stay broad:** Treatment markers show the planned schedule. Symptom bands are broad possibilities, not measured risk curves; serious symptoms can occur at any time. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00722 v2 · official regimen ↗️

> **There are no guaranteed "safe days".** Fever, shivers or suddenly feeling unwell require immediate contact. Serious symptoms override the timeline or diary.

Read the whole cycle as text

Day 1–21: The current treatment cycle

All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump. Your team confirms treatment dates and checks. Serious symptoms need advice on any day.

PRACTICAL TREATMENT GUIDE

Your practical treatment plan

All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump.



STEP 1

Before the first visit

Confirm blood tests, heart assessment, active medicines, route and the consent discussion.



STEP 2

Plan venous access

A cannula, PICC or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.



STEP 3

Infusion day

Trained staff check and access the line, give the prescribed premedicines and infusions, and monitor for reactions. Tell the nurse immediately about pain, burning, rash or breathlessness.



STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

First visits can take longer because of loading doses, premedicines and observation. Ask the unit about expected time, transport and any escort needed after sedation or line insertion.

Open practical treatment guide · Printable practical guide

Sources: NCCP 00722 v2 · official regimen · eviQ · implanted port patient information · eviQ · PICC patient information. Source reconciled; independent clinical, pharmacy and patient/PPI review pending.

MAKE THE PLAN PRACTICAL

Support around the medicines

1 The correct supportive prescription

Follow the script for the active medicine, route and treatment day. Do not carry chemotherapy anti-sickness medicines automatically into an antibody-only phase.

2 Tests before treatment

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.

3 Keep heart checks on the plan

An echo every 12 weeks is separate from blood tests or cancer imaging. Ask who books and reviews it.

4 **Plan for symptoms**

Keep your agreed contact numbers and instructions. New symptoms can need assessment before a routine appointment.

Sources: nccp · hse

TREATMENT REFERENCE · 01

Why this treatment?

TCHP treats higher-risk HER2-positive breast cancer before surgery using two chemotherapy medicines and two HER2-targeted antibodies. It avoids an anthracycline. Cancer stage, lymph nodes, fitness, heart and kidney function and expected side effects guide the choice. Surgery findings determine the next treatment.

Why my team might recommend it

TCHP treats higher-risk HER2-positive breast cancer before surgery using two chemotherapy medicines and two HER2-targeted antibodies. It avoids an anthracycline. Cancer stage, lymph nodes, fitness, heart and kidney function and expected side effects guide the choice. Surgery findings determine the next treatment.

Sources: nccp

What could change the choice?

Ask which cancer features, earlier treatments and health factors led to this recommendation, which alternatives were considered and when the decision will be reviewed.

Sources: nccp

TREATMENT REFERENCE · 02

Your treatment and the next step

Treat HER2-positive early breast cancer as part of a plan intended to reduce the risk of it returning.

Schedule

All four medicines are given by infusion on Day 1 of a 21-day cycle, for six cycles before surgery. Surgery findings guide the postoperative treatment. There is no routine take-home chemotherapy pump.

Sources: nccp

After this phase

After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

Sources: nccp

TREATMENT REFERENCE · 03

When scans, heart checks and bloods happen

Agree who books each test, who reviews the result and what happens if a check is overdue.

Blood tests

FBC, kidney and liver profile at baseline and before every 21-day cycle. The team reviews the necessary results before a planned dose.

Sources: nccp

Heart ultrasound and ECG

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

Sources: nccp · herceptin · esc

Scans and tumour markers

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine CT, PET or tumour-marker testing is not the usual screening plan for someone without symptoms.

Sources: early

Bone health and endocrine treatment

If the cancer is also hormone-receptor positive, endocrine treatment may form part of the wider plan. Bone-density testing is linked to aromatase inhibitors, ovarian suppression and fracture risk; it is not automatically required because a HER2 medicine is prescribed. Agree baseline DEXA where indicated and individual follow-up.

Sources: bone

TREATMENT REFERENCE · 04

How surgery and radiotherapy fit

Systemic treatment comes before surgery. Lumpectomy or mastectomy is chosen with the surgical team. Radiotherapy, if indicated, follows surgery and recovery; it may overlap postoperative HER2 treatment when both teams agree. It is not automatically concurrent with the chemotherapy phase.

The treatment relationship

Systemic treatment comes before surgery. Lumpectomy or mastectomy is chosen with the surgical team. Radiotherapy, if indicated, follows surgery and recovery; it may overlap postoperative HER2 treatment when both teams agree. It is not automatically concurrent with the chemotherapy phase.

Sources: early

What happens next?

After surgery, the pathology helps select the next HER2 medicine. The plan usually aims to complete about one year of HER2 treatment from the first dose. Residual invasive disease may change the treatment. Blood tests and supportive medicines must follow the selected postoperative regimen.

Sources: nccp

PREPARE FOR THE CONVERSATION

Questions worth asking

Who arranges my tests?

Ask the team who books each test, who checks the result and who to contact if an appointment is overdue. This map does not book appointments.

What if I become unwell between visits?

Use the urgent contact plan for sudden illness, fever, chest symptoms, severe diarrhoea or poor fluid intake. Do not wait for a scheduled blood test or scan.

Why was this treatment chosen?

TCHP treats higher-risk HER2-positive breast cancer before surgery using two chemotherapy medicines and two HER2-targeted antibodies. It avoids an anthracycline. Cancer stage, lymph nodes, fitness, heart and kidney function and expected side effects guide the choice. Surgery findings determine the next treatment.

When is my next heart scan?

Check heart pumping function before starting HER2 treatment. Repeat echo or MUGA every 12 weeks: around cycle 5, then cycle 9 if 21-day cycles stay on time. Earlier checks are needed for symptoms or abnormal findings. Count elapsed time from the first HER2 dose; changing phase or route does not restart the clock.

How do other treatments fit?

Systemic treatment comes before surgery. Lumpectomy or mastectomy is chosen with the surgical team. Radiotherapy, if indicated, follows surgery and recovery; it may overlap postoperative HER2 treatment when both teams agree. It is not automatically concurrent with the chemotherapy phase.

Do I need regular scans or markers?

Baseline breast and nodal imaging, with staging tests appropriate to the diagnosis. Review response during preoperative treatment and before surgery. After local treatment, arrange breast surveillance; routine CT, PET or tumour-marker testing is not the usual screening plan for someone without symptoms.

> **Try explaining the plan back:** Which medicines are active now, which days do you receive or take them, and who is booking and reviewing your next tests?

SOURCE TRANSPARENCY

Where this information comes from

Regimen schedule: NCCP 00722 v2. Medicine explanations and safety information: sources below. The timeline is an educational illustration, not a measured risk curve or an individual prediction.

- NCCP 00722 v2 · official regimen
- ESMO 2024 early breast cancer guideline · treatment and follow-up
- ESC 2022 cardio-oncology guideline · cardiac surveillance and risk
- HSE · chemotherapy side effects and urgent advice
- NICE NG101 · bone health and follow-up
- NOGG · section 7, aromatase inhibitors and fracture risk
- eviQ · implanted port patient information
- eviQ · PICC patient information
- NCCP NCIS antiemetics v8 (2025), Table 1
- ASCO · biomarkers in metastatic breast cancer
- ESMO metastatic breast cancer guideline · 2026
- EMA Herceptin product information
- EMA Perjeta product information

SACTCheck · Developed by Dr Paul O'Brien · Sources checked 2026-10-21.
This information supports your treatment discussion. Your oncology team confirms your personal treatment and urgent-care plan.

Written source addresses

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https://academic.oup.com/eurheartj/article/43/41/4229/6673995

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